

But you have two moving objects: the wheel and the ball. You mentioned the wheel, what about the ball?

The way it worked was that you would hit the toe switch when double zero passed the mark, and you would hit the switch again the next time it passed the mark. So now, the computer knows how fast the rotor is going. The ball is spun after the rotor measurements.

Oh, I see, there is no need for simultaneous measurements. First you measure the speed of the wheel, and then you measure the speed of the ball, each time taking two points.

Exactly, there are four clicks: the first two tell you the velocity of the rotor, and the second two tell you the velocity of the ball. My idea was for the computer signal to be the tones in the octave—each tone corresponding to an octant of the wheel. Shannon came up with the beautiful idea to design the program so that it cycled through the tones, providing the best estimate at each moment. When you clicked the fourth time, the tones would stop. The last tone you heard was the octant to bet on. What was nice about that approach was there was no compute time. The computer was continuously calculating the best estimate, and when you clicked the last time, the computation was over.

So as time is going by, the estimate is constantly changing.

Exactly, each tone tells you what the current estimation is. The program was designed to give you the estimate, assuming the last click occurred at that exact moment. Once we heard the last tone, we would bet on all five numbers in that octant.

When did you actually apply your roulette betting system in a casino?

Shannon and I and our wives went on a trip to Vegas in August 1961. We had to wire ourselves up.

Who wore the computer and who wore radio receiver?

Shannon was better at estimating the ball position than I was, so he was the timer and wore the computer. He stood next to the wheel and wrote down numbers just like any silly system player.

As a diversion.